CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 14: Equation of a Line

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3,3)$, $B(0,-2)$, $C(-4,-2)$ and $D(-1,3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC. ICSE 2025, 5 marks

2. Triangle ABC has vertices $A(1,-1)$, $B(0,4)$ and $C(-6,4)$, and D is the mid-point of BC. Find (a) the coordinates of D (b) the equation of the median AD. ICSE 2023, 3 marks

3. $A(2,5)$, $B(-1,2)$ and $C(5,8)$ are the vertices of a triangle. M is the point on AB with $AM : MB = 1 : 2$. Find M, and then the equation of the line through C and M. ICSE 2018, 4 marks

2 Lines parallel or perpendicular to a given line

4. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ ICSE 2025, 1 mark

5. The line PQ meets the positive x-axis at P, 5 units from the origin, and the negative y-axis at Q, 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ. Find (a) the coordinates of P and Q (b) the equation of RS. ICSE 2025, 3 marks

6. (a) The lines $kx - y + 4 = 0$ and $2y = 6x + 7$ are perpendicular. Find k. (b) Find the equation of the line parallel to $2y = 6x + 7$ that passes through $(-1,1)$. ICSE 2024, 3 marks

7. The lines $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular. Find p. Then find the equation of the line through $(-2,-1)$ parallel to $6x - py + 7 = 0$. ICSE 2020, 3 marks

8. The vertices of $\triangle ABC$ are $A(3,8)$, $B(-1,2)$ and $C(6,-6)$. Find (a) the slope of BC (b) the equation of the line through A perpendicular to BC. ICSE 2019, 3 marks

9. The lines $3x - 5y = 7$ and $4x + ay + 9 = 0$ are perpendicular. Find a. ICSE 2018, 3 marks

10. $A(-1,3)$, $B(4,2)$ and $C(3,-2)$ are the vertices of a triangle. Find (a) the centroid G (b) the equation of the line through G parallel to AC. ICSE 2017, 3 marks

3 Inclination and slope

11. The points $A(x,y)$, $B(3,-2)$ and $C(4,-5)$ are collinear. Write y in terms of x. (a) $3x - 11$ (b) $11 - 3x$ (c) $3x - 7$ (d) $7 - 3x$ ICSE 2024, 1 mark

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu